



Future of Hybrid Cloud

Jason Payne
CTO, US Federal

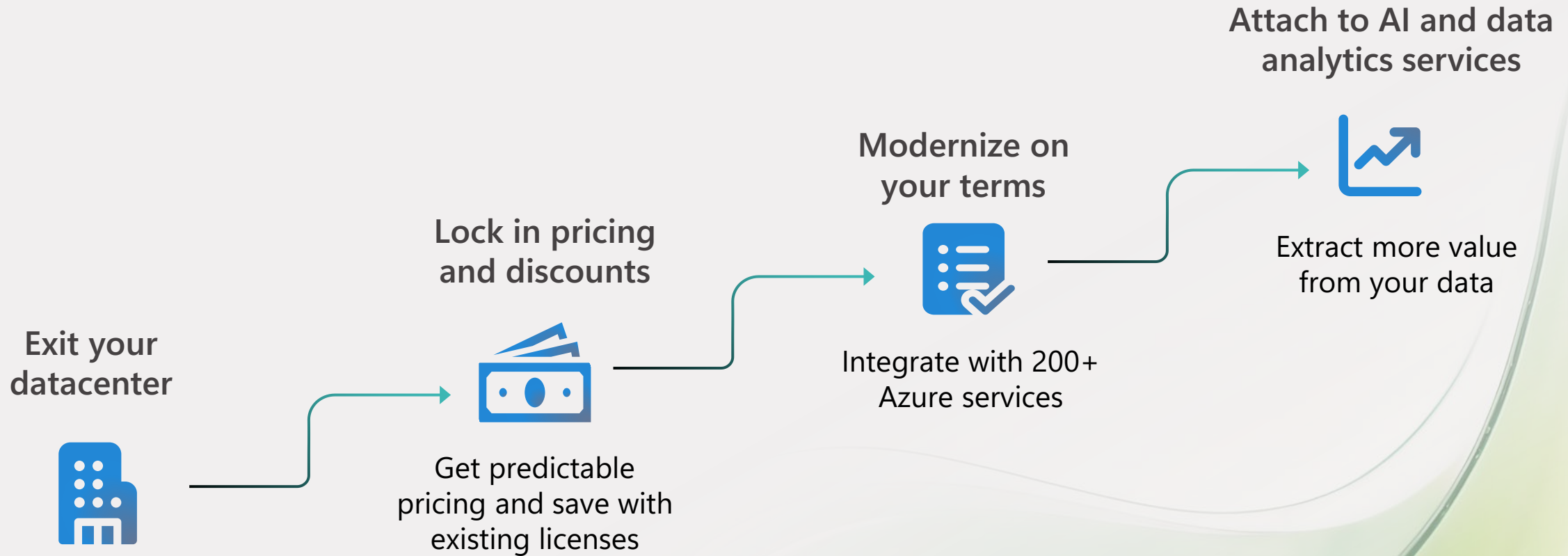
By 2028, at least

15%

of day-to-day work decisions will be made autonomously through agentic AI, up from 0% in 2024.

Source: [Top Technology Trends in Government for 2025](#), October 2024

Migrate to modernize - the hybrid reality



Government challenges specific to hybrid cloud

Complexity

"I need to have health visibility in a single pane of glass to all my existing and future infrastructure and applications."

Regulation

"Our DB layer must remain on-premises due to sensitive patient data and data availability needs."

Compliance

"I need to manage security and incident management across my public cloud and datacenter assets."

Latency

"We can't take a dependency on the internet. If we lose connectivity, we still want to be able to access the data."

Inconsistency

"I want my on-prem skills to work in the cloud, and my cloud skills to work on-prem."

Legacy

"Our older systems take too much maintenance. We want evergreen technology and to pay for it like a utility."

Four government use cases enabled by hybrid cloud



Be AI-ready within government agencies

- Use analytics and AI to augment existing mission applications.
- Implement IoT and machine learning to predict and prevent digital infrastructure failures.
- Leverage digital platforms to automate and optimize business processes.



Mitigate risks with built-in security and compliance

- Automate threat detection and response by implementing AI-driven security systems.
- Deploy comprehensive compliance management solutions that automate the tracking and reporting of regulatory requirements.
- Utilize advanced encryption methods and software-defined sovereign controls to protect sensitive government data.



Purpose-built for all workloads while supporting regulatory requirements

- Implement online digital service portals to streamline processes, reduce wait times for the public, and ensure uptime and availability.
- Use IoT sensors and data analytics to monitor and manage public infrastructure at scale.
- Deploy intelligent systems that optimize logistics, improve efficiency, and provide real-time information to mission owners, citizens, and soldiers.



Cloud agility for agencies anywhere

- Provide remote access for government employees to securely access critical applications and data from any location, enhancing flexibility and continuity of operations.
- Enable tools that facilitate seamless collaboration across departments and agencies, regardless of location.
- Allow for rapid resource scaling to meet demands, ensuring that services remain responsive and effective during peak times or emergencies.

Be AI-ready within government agencies

Migration to Azure accelerates innovation with AI by using co-location, infusing Azure AI Service and Copilots, and embracing Responsible AI

Improve AI performance and accelerate innovation across government with purpose-built AI supercomputing infrastructure.

Unify data, apps, and infrastructure in the cloud for optimal latency and cost-savings to effectively infuse AI into agencies.

Secure and responsible AI throughout the machine learning development lifecycle to build fair, explainable applications that earn the trust of workers and the public.

70%

By 2027, GenAI tools will be used to explain legacy business applications and create appropriate replacements, reducing modernization costs by 70%.



Mitigate risks with built-in security from code to cloud

Secure from code to cloud with foundational software defined federal controls and compliance

Enhance team impact and efficiency by protecting with Microsoft Defender, Sentinel, Purview, etc. tailored for government operations.

Transform data into actionable insights with Sentinel on the Azure portal, providing AI-driven Security Information and Event Management (SIEM) tailored for government agencies.

Create compliant architectures to meet federal compliance requirements with codified architectures, workload templates, and tooling.

80%

lowered risk of data breach from unsecured infrastructure



Enabling Hybrid Migration and Modernization

Scenario and workload support



Infrastructure and database migration

for Windows Server, SQL Server, Linux, OSS-databases, DevTest, migrating to VMware to Azure VMware Solution, hybrid deployment with Arc-enabled servers and data



Migrate enterprise apps

for .NET and Java apps migrating to Azure App Service



Virtual desktop infrastructure

for Windows 10 and 11 based virtual desktops migrating to Azure Virtual Desktop (Including VMware and Citrix solutions)



SAP

for SAP native environments, SAP HANA migration, greenfield SAP deployments and SAP RISE



Oracle

for Oracle Database migrating to Azure VMs or to other database platforms, Oracle app migrations (JD Edwards, PeopleSoft, Siebel, and others) and Oracle Exadata migrations.



High performance compute and AI infrastructure

including simulation, modelling, visualization and AI training/inferencing at large scale



Mainframe and midrange migration

for applications running on IBM z/Series, IBM i, Power Systems or others

Azure Migration & Modernize



Comprehensive Support:

Extensive resources, tools, and expert guidance for smooth migration and modernization.

Specialized Partners: Over 700 partners provide expertise and support, helping to offset project costs.

Security and Compliance: Built-in security guidance ensures adherence to best practices and compliance requirements.

Azure VMware Solution

Your fastest path to the cloud

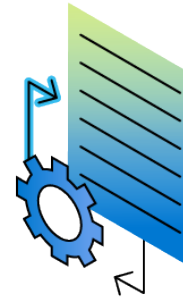
Accelerate, modernize, simplify

1



Migrate fast
without refactoring

2



Innovate applications
at your own pace

3



Achieve better
performance and security

4



Leverage existing
Microsoft licenses

What you get with Azure VMware Solution

Fastest migration option

- ✓ Migrate VMware workloads “as-is”
- ✓ No refactoring or application modifications required
- ✓ Support from expert migration partners

Familiar, consistent VMware technology

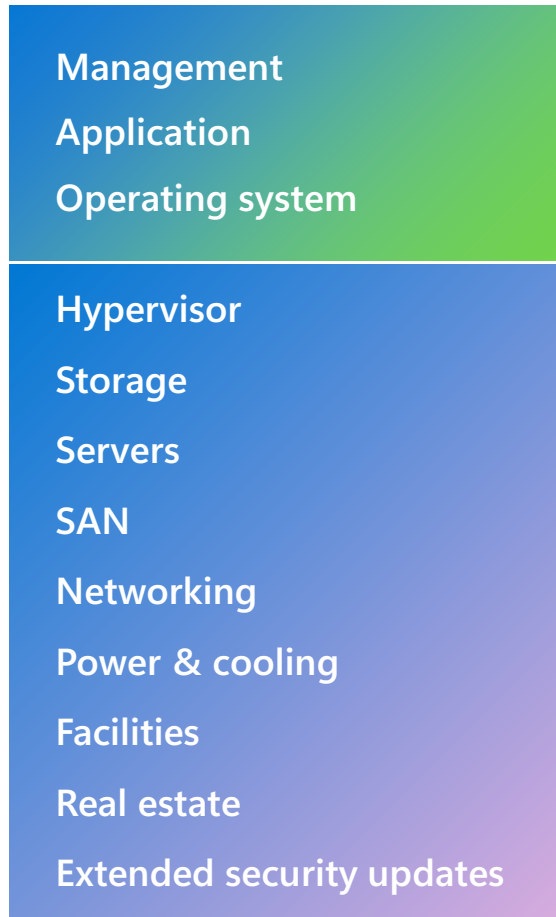
- ✓ Use VMware tech stack and VCF subscriptions on Azure infrastructure
- ✓ Included technology: vSphere, vSAN, NSX, HCX, and Aria
- ✓ Continue to leverage existing VMware expertise

Fully-managed, first-party Azure service

- ✓ Azure provides support, licensing, and billing
- ✓ Manage VMware resources through the Azure portal
- ✓ Add capacity in minutes “VMware as a Service”

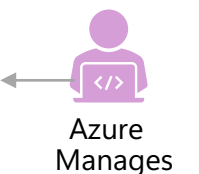
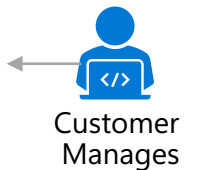
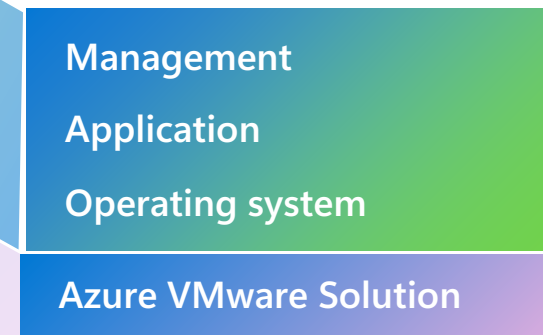
Free up resources

VMware on premises



Allow resources to spend more time focusing on tasks that add higher value to your organization.

VMware in the cloud



Microsoft Azure



Single control plane with Azure Arc

Infrastructure

Connect and operate hybrid resources as native Azure resources

Azure Arc-enabled infrastructure

Services

Deploy and run Azure services outside of Azure while still operating it from Azure

Azure Arc-enabled services



Arc Server



K8s



Windows



SQL Server



SQL DB



PostgreSQL



Web Apps



Functions



Logic Apps



Machine Learning



Multi-cloud

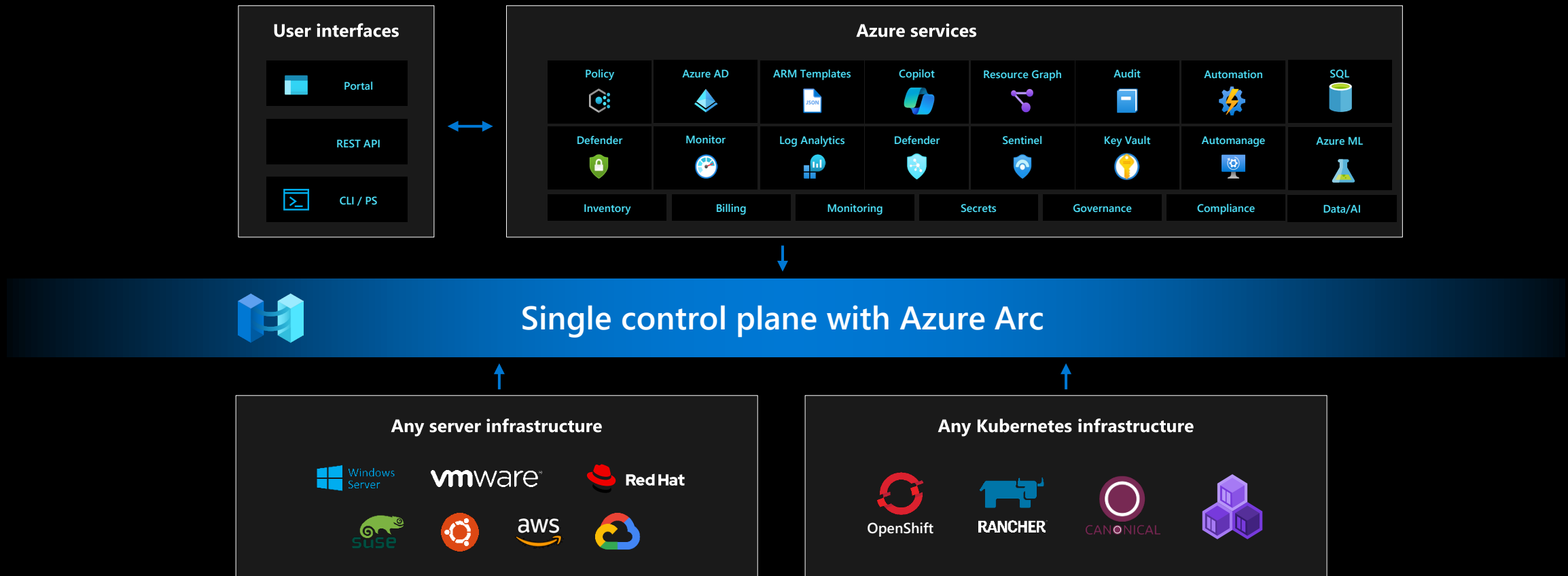


Datacenter



Edge

Azure Arc: extending Azure management & security to any infrastructure



Manage your private, public and edge environment in a consistent way

Azure Virtual Desktop is a cloud VDI solution designed to meet the challenges of remote work

Enable a secure,
remote desktop
experience
from anywhere



Access Windows 11 and Windows 10 from anywhere



Maintain full control over configuration and management



Get the security and reliability of Microsoft Azure



Optimize cost with multi-session and pay for only what you use

Access Windows 11 and Windows 10 from anywhere



Provide access to Windows 11 and Windows 10
on a variety of devices



Deliver a seamless experience on
Microsoft Office and Microsoft Teams



Personalize the Windows 11 and Windows 10
virtual experience with roaming user profiles

Maintain full control over configuration and management



Enable GPU accelerated app rendering and encoding on Azure Virtual Desktop session hosts



Empower IT to specify how desktops are distributed across VMs



Simplify the delivery of company-specific apps to employees with capabilities for external users

Get the security and reliability of Microsoft Azure



Use Microsoft Entra ID and Microsoft Entra multifactor authentication (MFA) to secure your virtual desktops



Deploy your virtual infrastructure in Azure datacenters around the world

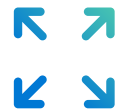


Provide management and customization options for IT by leveraging Azure services such as Microsoft Azure Monitor

**Optimize cost
with exclusive
Windows 11 and
Windows 10
multi-session and
pay for only what
you use**



Run multiple desktop sessions on a single VM with exclusive Windows 11 and Windows 10 multi-session

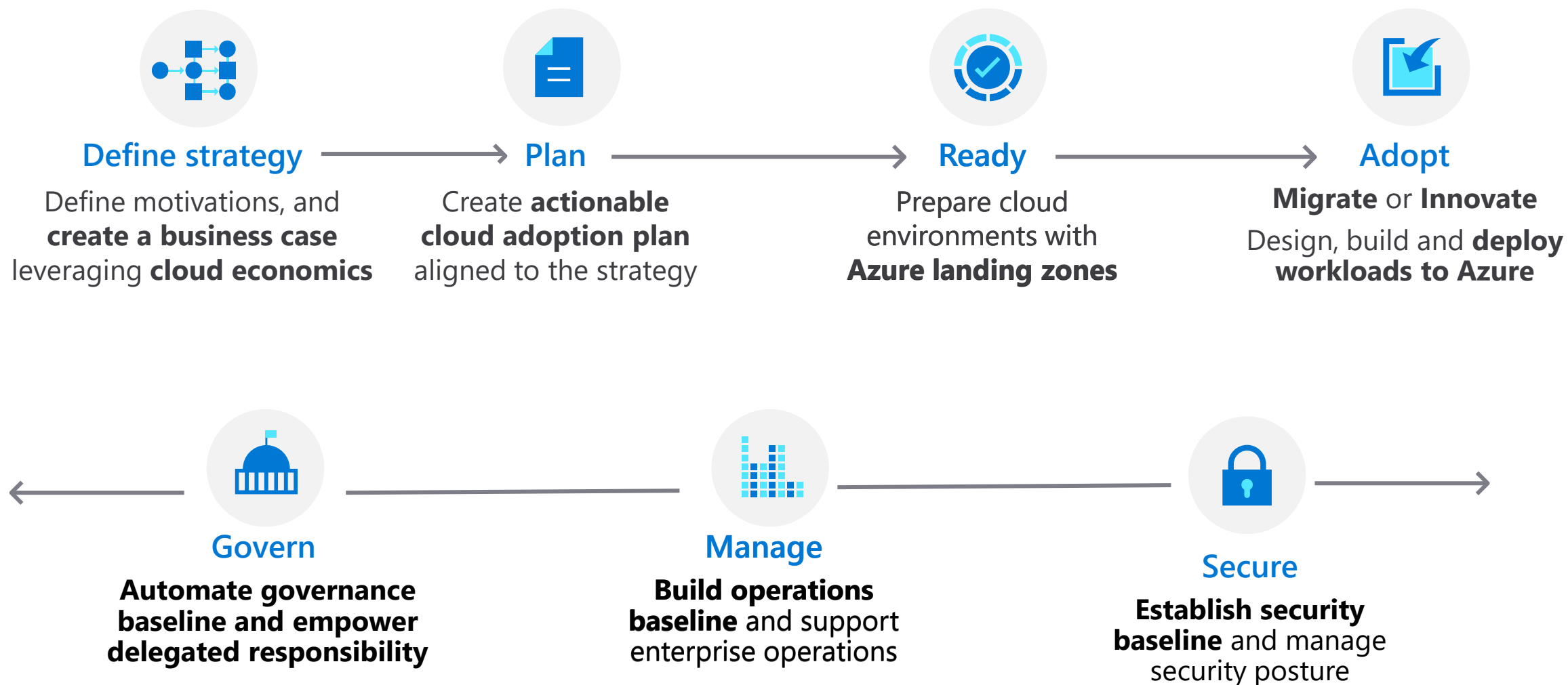


Optimize deployment costs and scale session host VMs with autoscale

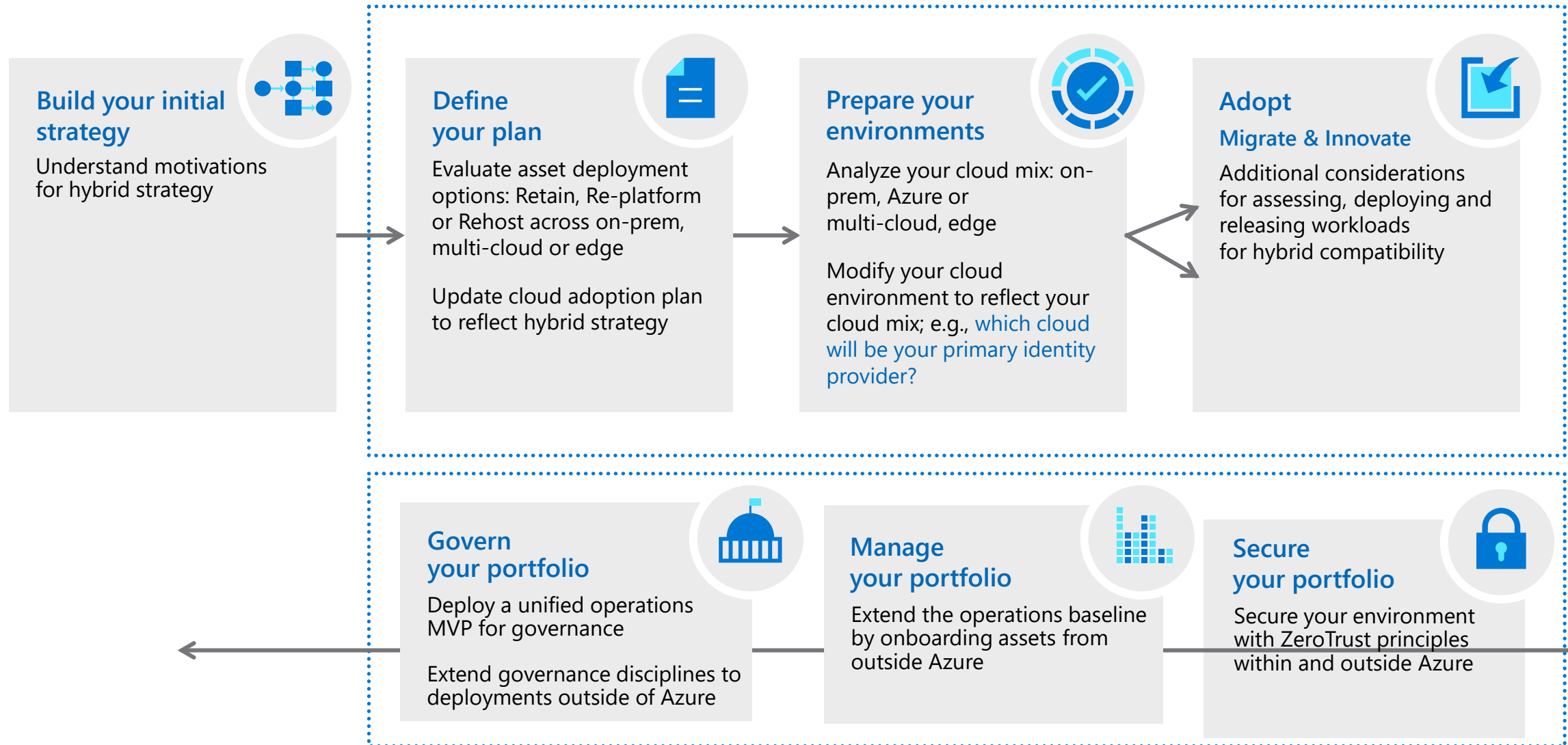


Use existing Windows and Microsoft 365 licenses to access Windows 11 and Windows 10 and Microsoft Office

Microsoft Cloud Adoption Framework for Azure



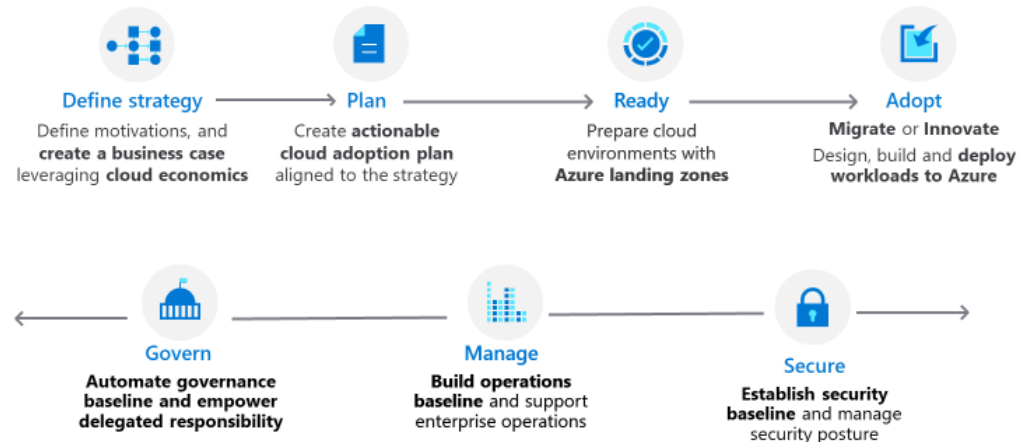
Hybrid and multicloud adoption journey



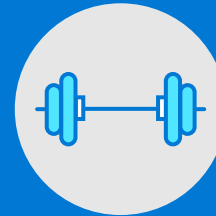
Take action and implement throughout the journey (1 of 2)

Start with Cloud adoption framework to guide your journey

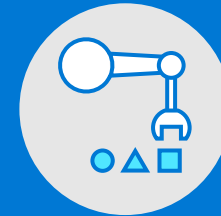
Microsoft Cloud Adoption Framework for Azure



Hybrid and multicloud: Supporting assets and resources



Build skills across your team with Microsoft Learn



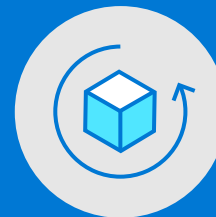
Apply best practices to rapidly onboard



Accelerate deployment with reference architectures



Deep knowledge on products with Docs



Optimize workloads with Azure Well-Architected

<https://aka.ms/adopt/hybrid>

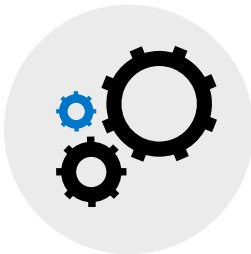
Well-Architected hybrid workloads

- Hybrid workloads are complex with applications running across on-premises, multicloud, and the edge.
- Use Cloud Adoption Framework to establish the environment, then migrate, govern, and manage your complete portfolio of workloads.
- Use Azure Well-Architected Framework to review & modify high priority or complex workloads that may benefit from architectural improvements.
- The [Azure Well-Architected Framework](#) provides guidance for hybrid workloads across pillars of cost optimization, operations excellence, performance efficiency, reliability and security.

Cost
Optimization



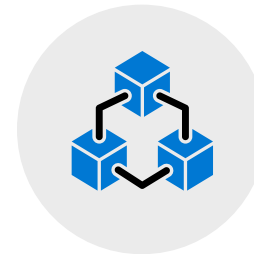
Operational
Excellence



Performance
Efficiency



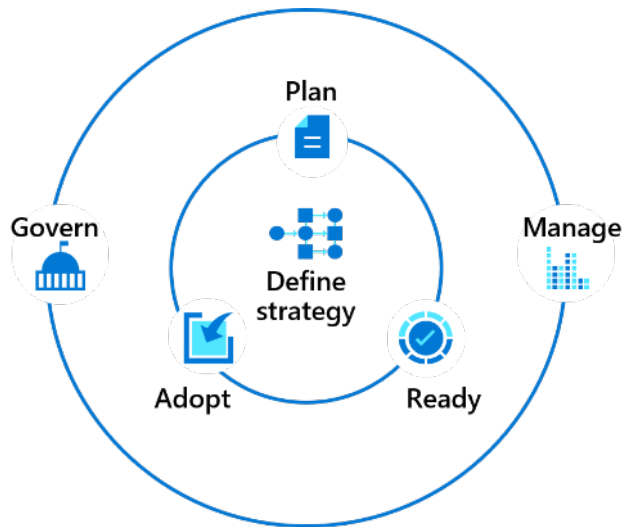
Reliability



Security

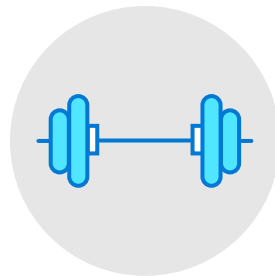


Complete guidance for hybrid and multicloud approach



Cloud Adoption Framework

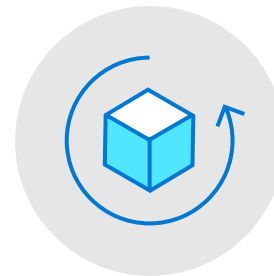
Guide your cloud journey across people, process and technology



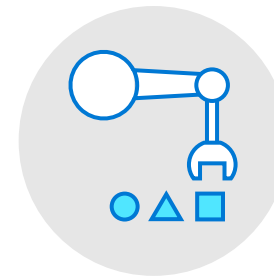
Build skills across your team with **Microsoft Learn**



Accelerate deployment with **Reference Architectures**



Optimize workloads with **Azure Well-Architected**



Apply **best practices** to rapidly onboard



Review **technical documentation** on featured products

Start with the cloud adoption framework to guide your cloud journey and build on it using the hybrid adoption scenario guidance

<https://aka.ms/adopt/hybrid>